CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
DEPARTMENT of COMPUTER SCIENCE AND ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION
(Inline with AICTE Model Curriculum with effect from AY 2026-27)
(R22A Regulation)

SEMESTER - V

S. No	Course Code	Title of the Course	Scheme of Instruction			Scheme of Examination			Credits
			Hours per Week			Duration of SEE in Hours	Maximum Marks		
			L	T	P/D		CIE	SEE	
THEORY									
1.	22ITC10N	Computer Networks	3	-	-	3	40	60	3
2.	22CSC15N	Operating Systems	3	-	-	3	40	60	3
3.	22CSC21N	Software Engineering	3	-	-	3	40	60	3
4.	22CSC59	Fundamentals of Data Science	3	-	-	3	40	60	3
5.	22CSExx	Professional Elective Course-I	3	-	-	3	40	60	3
6.	22xxxxx	Open Elective Course – I	3	-	-	3	40	60	3
PRACTICAL									
7.	22ITC11N	Computer Networks Lab	-	-	3	3	50	50	1.5
8.	22CSC18N	Operating Systems Lab	-	-	3	3	50	50	1.5
9.	22CSC49	Software Engineering Lab	-	-	2	3	50	50	1
Total			**18**	**-**	**8**	**-**	**390**	**510**	**22**

L: Lecture **T: Tutorial** **P: Practical**
CIE: Continuous Internal Evaluation **SEE-Semester End Examination**

Professional Elective – I	
22CSE01	Computer Graphics and Multimedia
22CSE03N	Optimization Techniques
22CSE17	Mobile Application Development
22CSE19	Big Data Analytics
22CAE03N	Image Processing

Open Elective - I	
22BTO01	Biology For Engineers
22CEO02	Disaster Risk Reduction and Management
22CHO04	Environmental and Sustainable Development
22MEO01	Principles of Design Thinking
22ECO05	Principles of Embedded Systems

22ITC10N

COMPUTER NETWORKS

Instruction	3 L Hours per Week
Duration of SEE	3 Hours
SEE	60 Marks
CIE	40 Marks
Credits	3

COURSE OBJECTIVES: This Course aims to

1. Understand the basics of Layering Concepts, Physical layer, data transmission, transmission media.
2. Demonstrate the state-of-the-art knowledge on data link layer concepts.
3. Distinguish the different types of routing algorithms and network layer in the Internet.
4. Introduce Transport Layer basics, UDP and TCP Protocols.
5. Know the concepts of Application Layer Protocols.

COURSE OUTCOMES: After the completion of this course, the student will be able to

1. Explain the functions of each layer in the OSI, TCP/IP reference models and demonstrate the concepts of Physical Layer.
2. Analyse the Data Link Layer protocols and MAC mechanisms.
3. Evaluate the Routing algorithms and the IP Protocols.
4. Illustrate the functions and performance of Internet Transport Protocols TCP and UDP.
5. Explore the various Application layer protocols.

CO-PO Articulation Matrix

PO/PSO CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO 1	1	2	1	2	1	-	-	-	-	-	-	2	3	2
CO 2	3	3	3	2	1	-	-	-	-	-	-	3	3	2
CO 3	3	3	2	2	3	-	-	-	-	-	-	3	3	2
CO 4	3	3	2	2	2	-	-	-	-	-	-	3	3	3
CO 5	2	2	1	2	3	3	3	2	3	3	3	3	3	2

1 - Slightly, 2 - Moderately, 3 – Substantially

UNIT-I

Introduction: Network Hardware, Network Topologies, Reference Models- The OSI Reference Model- the TCP/IP Reference Model – A Comparison of the OSI and TCP/IP Reference, Packet Switching, Circuit Switching and Virtual Circuit Switching. Physical Layer: Guided Transmission media, Twisted Pairs, Coaxial Cable, Fiber Optics, Wireless transmission.

UNIT-II

Data Link Layer: Design Issues, Error Detection and Correction, Elementary Data Link Protocols: Simplex Protocol, A Simplex Stop and Wait Protocol for an Error-free channel, Sliding Window Protocols, Go-Back-N, Selective Repeat. Medium Access Sub Layer: The Channel allocation problem, Multiple Access Protocols: ALOHA, Carrier Sense Multiple Access Protocols.

UNIT- III

Network Layer: Design Issues, Routing algorithms: The Optimality Principle, Shortest Path Algorithm, Flooding, Distance Vector Routing, Link State Routing, OSPF, BGP, Quality of Service, The Network layer in the Internet- The IP Version 4 Protocol, IP Addresses, IP Version 6.

UNIT-IV

Transport Layer: Transport Service, Berkeley Sockets, Elements of Transport Protocols, **The Internet Transport Protocols:** Introduction to UDP, The TCP Protocol, The TCP Segment Header, The TCP Connection Establishment, TCP Connection Release, TCP Sliding Window, TCP Timer

Management, TCP Flow Control, Congestion Control.

UNIT-V

Application Layer: DNS, The Domain Name System, The DNS Name Space, Domain Resource Records, Name Servers. **Electronic MAIL:** Architecture and Services, The User Agent, Message Formats, Message Transfer, Final Delivery, SMTP, FTP, TELNET, SNMP.

TEXT BOOKS:

1. Andrew S. Tanenbaum, David J. Wetherall, "Computer Networks", 5th Edition, Pearson Education, 2014.

SUGGESTED READING:

1. Andrew S. Tanenbaum, David J. Wetherall, "Computer Networks", 5th Edition, Pearson Education, 2021.
2. Chwan-Hwa (John) Wu, J. David Irwin, "Introduction to Computer Networks and Cyber Security", CRC Press, 2013.
3. W. Richard Stevens, "Unix Network Programming", Prentice Hall/Pearson Education, 2009.
4. James F. Kurose and Keith W. Ross, "Computer Networking: A Top-Down Approach Featuring the Internet", 5th Edition, Addison-Wesley, 2012.
5. Larry L. Peterson and Bruce S. Davie "Computer Networks: A Systems Approach", 5e, 2018
6. Behrouz A. Forouzan "Data Communications and Networking", Fourth Edition,2007.

ONLINE RESOURCES:

1. https://nptel.ac.in/courses/117105148,title of the course
2. Computer Networks - Books, Notes, Tests 2025-2026 Syllabus
3. IEEE Transactions on Networking | IEEE Communications Society
4. Web Resources for Computer Networks, 5 (vu.nl)

22CSC15N

OPERATING SYSTEMS

Instruction	3 L Hours per Week
Duration of SEE	3 Hours
SEE	60 Marks
CIE	40 Marks
Credits	3

PREREQUISITE: Computer Architecture and Programming Fundamentals.

COURSE OBJECTIVES: This course aims to
1. Understand the basic concepts and design of an operating system.
2. Interpret the structure and organization of the file system.
3. Learn Inter Process Communication mechanisms and memory management approaches.
4. Explore cloud infrastructures and technologies.

COURSE OUTCOMES: After the completion of this course, the student will be able to
1. Understand the basics of Operating systems and its major components.
2. Illustrate the concepts related to process management.
3. Distinguish various memory management techniques.
4. Apply concepts of process synchronization and deadlocks to a given situation.
5. Evaluate various file allocation methods and security as well as recovery features in the designing Operating system.

CO-PO Articulation Matrix

PO/PSO CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO 1	2	1	1	-	-	-	-	-	-	-	-	2	1	1
CO 2	3	3	-	3	1	-	-	-	-	-	-	2	2	2
CO 3	3	3	2	1	1	-	-	-	-	-	-	2	2	2
CO 4	3	3	1	3	-	-	-	-	-	-	-	2	2	2
CO 5	3	3	2	3	1	-	-	-	-	-	-	2	2	2

1 - Slightly, 2 - Moderately, 3 – Substantially

UNIT-I

Introduction to Operating Systems: Computer System overview, Components of a computer system, functions of OS, Examples and different types of OS *(RTOS vs. desktop vs. mobile etc.),* OS distributions and versions. **OS architectures:** Micro-kernel, Layered, Kernel Approaches and examples, Linux/Unix OS Design and architecture overview.

UNIT-II

Process management: Program vs. process, process states, Process Control Block (PCB), OS services and system calls *(fork, wait, exec, getpid, getppid etc.),* system calls vs. System programs, Process scheduling- Process context switching, scheduling algorithms, scheduling criteria. Process management in Unix/Linux. **Inter Process Communication:** Linux IPC Mechanisms, RPC, RPC exception handling and Security issues.

UNIT-III

Memory Management: Memory view of a process, Process memory usage requirements, virtual and physical memory related system calls *(mmap, munmap, sbrk, mprotect).* Address translation mechanisms static mapping, segmentation, paging, page faults, page replacement algorithms, page sharing, read/write permissions, swapping. Memory management in Linux/Unix. **Secondary Memory Management:** Disk structure, disk scheduling, disk management, buffering, swap space management.

UNIT-IV

Concurrency and Synchronization: Introduction to threads, benefits, types and thread APIs, Synchronization, issues, hardware and software solutions for synchronization, Classical problems of synchronization. **Deadlocks:** Introduction, necessary conditions for deadlock occurrence, RAG, deadlock handling mechanisms - prevention, avoidance and recovery.

UNIT-V

File Systems: File concepts, file types, allocation and organization methods, file handling system calls, File system metadata, directory structure, caching optimizations, File Systems case study. **OS Security and Defence:** Types of threats in OS, basic security mechanisms, malware taxonomy, viruses, worms, and rootkits, logging, auditing, and recovery.

TEXT BOOKS:

1. Galvin, Silberschatz, "Operating system Concepts", 10th Edition, John Wiley & Sons, 2018.
2. William Stallings, "Operating Systems Internals and Design Principles" Pearson Edition, 2012.

SUGGESTED READING:

1. Ekta Walia Khanna, "Operating System Concepts", 2nd Edition, Publishing House, 2019.
2. W. Richard Stevens, Stephen A. Rago, "Advanced Programming in the UNIX® Environment", 3rd Edition, Pearson Education India; 2013.
3. Maurice J. Bach, "Design of the UNIX Operating System", 1st Edition, Pearson Education India, 2015.

ONLINE RESOURCES:

1. Remzi H. Arpaci-Dusseau and Andrea C. , "Three Easy Pieces", Arpaci-Dusseau Arpaci-Dusseau Books, LLC https://pages.cs.wisc.edu/~remzi/OSTEP/ (online version).
2. Frans Kaashoek, Robert Morris, and Russ Cox, Xv6, a simple Unix-like teaching operating system [T4-R] https://github.com/mit-pdos/xv6-riscv (RISC-V version) [T4-X] https://github.com/mit-pdos/xv6-public (x86 version).

22CSC21N

SOFTWARE ENGINEERING

Instruction	3 L Hours per week
Duration of SEE	3 Hours
SEE	60 Marks
CIE	40 Marks
Credits	3

PREREQUISITE: Object-oriented programming, Programming for problem-solving, database management systems.

COURSE OBJECTIVES: This course aims to

1. Understand the Software Engineering Practice and Process Models.
2. Understand Design Engineering and Project Management in Software Development.
3. Understand the importance of testing in software development and study various testing strategies and software quality metrics.

COURSE OUTCOMES: After the completion of this course, the student will be able to

1. Acquire a working knowledge of software processes and models for each phase of software development.
2. Understand the agile Software process models and demonstrate the skills necessary to specify the requirements.
3. Recall the modelling concepts and estimate the cost of software using empirical models.
4. Enlist the design principles and construct a product using coding principles and standards.
5. Develop test cases and acquire skills necessary for independently developing a complete software project and estimate software quality.

CO-PO Articulation Matrix

PO/PSO CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO 1	3	3	2	1	3	3	-	3	3	1	3	1	-	3
CO 2	3	2	3	1	-	2	-	3	3	1	3	1	-	3
CO 3	2	3	3	3	3	1	-	3	3	3	3	1	-	3
CO 4	3	3	3	2	3	2	-	3	3	2	3	1	-	3
CO 5	3	3	1	2	2	1	-	3	3	2	-	1	-	3

1 - Slightly, 2 - Moderately, 3 – Substantially

UNIT - I

Introduction to Software Engineering: Software Engineering Practice, The Software Process, Software Engineering Practice Process Models: A Generic Process Model, Process assessment and Improvement, Prescriptive Process Models: Waterfall Model, Incremental Process Models, RAD Model, Evolutionary Process Models - Prototyping, The Spiral Model, Specialized Process Models.

UNIT - II

An Agile Development: Agility, Agile Process, and Agile Process Models- Extreme Programming (XP), Adaptive Software Development (ASD), Scrum, Dynamic Systems Development Method (DSDM), Feature Driven Development (FDD), Agile Modelling (AM), Requirement Engineering, Establishing the groundwork, Eliciting Requirements, Negotiating Requirements, and Validating Requirements. Software Requirements Analysis and Specification: Value of a Good SRS, Problem Analysis, Requirements Specification.

UNIT - III

Planning a software Project: Effort Estimation, Project Schedule and Staffing, Quality Planning, Risk Management, Estimation for Software Projects: Decomposition Techniques - Software Sizing, Problem-Based Estimation, An Example of LOC-Based Estimation, An Example of FP-Based Estimation, COCOMO Model

UNIT - IV

Function-oriented modelling, Design Concepts -Coupling, Cohesion, Flow-oriented modelling-DFDs with Examples, Software Architecture, A Brief Taxonomy of Architectural Styles, Component-Level Design: Definition, Basic Design Principles, Design Guidelines, Designing Traditional Components, Coding Principles and guidelines, Incremental Development of Code, Code Inspection – Planning.

UNIT - V

Testing - Testing Concepts, Testing Process, Testing Strategies: A Strategic approach to software testing, strategic issues, test strategies for Conventional Software, Validation Testing, System Testing, White Box Testing, Black Box. Automatic vs. Manual Testing, Software Review Techniques - Informal Reviews Formal Technical Reviews, Quality Concepts - What is Quality, Software Quality, Objectives, Software Quality Attributes (McCall's,HP)Deployment overview, Deployment planning, Deployment Rollback.

TEXT BOOKS:

1. Roger S. Pressman "Software Engineering: A practitioner's approach", McGraw Hill, 8th Edition, 2014.
2. Pankaj Jalote, "Software Engineering Precise Approach", Wiley Publishers, 2012

SUGGESTED READING:

1. Sommerville "Software Engineering", 10th Edition, Pearson, 2016.
2. Rajib Mal "Fundamental of Software Engineering", 4th Edition, PHI Learning, 2014.

ONLINE RESOURCES:

1. https://nptel.ac.in/courses/106101061/
2. Udemy:https://www.udemy.com/share/101BHy3@YYJn8BxwvS6cGfnCsiIlxyA-lUjwZmA2xN5WmMbd8hlGxwhc4N0DF7KaEOaz4eDnMg==/

22CSC59

FUNDAMENTALS OF DATA SCIENCE

Instruction	3 L Hours per Week
Duration of SEE	3 Hours
SEE	60 Marks
CIE	40 Marks
Credits	3

COURSE OBJECTIVES: This course aims to

1. Understanding the fundamental concepts and scope of Data Science.
2. Develop a strong foundation in statistical methods and data analytics techniques.
3. Understanding data manipulation and analysis using Python libraries.
4. Understanding data preprocessing techniques.
5. Familiarize with supervised and unsupervised machine learning techniques.

COURSE OUTCOMES: After the completion of this course, the student will be able to

1. Understand the fundamental concepts of Data Science and its interdisciplinary nature, including its relationship with statistics.
2. Apply statistical and analytical techniques such as descriptive, diagnostic, predictive, and prescriptive analytics.
3. Implement data manipulation and analysis tasks using NumPy and pandas.
4. Analyze and clean real-world datasets using various preprocessing, filtering, encoding, and hypothesis testing techniques.
5. Develop and evaluate machine learning models for regression and clustering tasks using supervised and unsupervised learning techniques.

CO-PO Articulation Matrix

PO/PSO CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO 1	3	3	2	3	3	-	1	-	-	-	2	3	3	3
CO 2	3	3	2	3	2	-	1	-	2	-	2	3	3	3
CO 3	3	3	3	3	3	-	1	-	1	-	2	3	3	2
CO 4	3	3	3	3	3	2	1	1	-	-	2	3	3	2
CO 5	3	3	3	3	3	2	1	-	-	-	3	3	3	2

1 - Slightly, 2 - Moderately, 3 – Substantially

UNIT–I

Introduction: What Is Data Science? Where Do We See Data Science? Finance, Public Policy, Politics, Healthcare, Urban Planning, Education, Libraries. How Does Data Science Relate to Other Fields: Data Science and Statistics, Data Science and Computer Science, Data Science and Engineering, Data Science and Business Analytics, Data Science, Social Science, and Computational Social Science, The Relationship between Data Science and Information Science: Information vs. Data Users in Information Science, Data Science in Information Schools (iSchools). Computational Thinking. Skills for Data Science, Tools for Data Science, Issues of Ethics, Bias, and Privacy in Data Science. **Data:** Introduction, Data Types: Structured Data, Unstructured Data, Challenges with Unstructured Data. **Data Collections:** Open Data, Social Media Data, Multimodal Data, Data Storage and Presentation. Data **Pre-processing:** Data Cleaning, Data Integration, Data Transformation, Data Reduction, Data Discretization.

UNIT– II

Techniques: Introduction: Data Analysis and Data Analytics, Descriptive Analysis: Variables, Frequency Distribution, Measures of Centrality, Dispersion of a Distribution. **Diagnostic Analytics:** Correlations, Predictive Analytics, Prescriptive Analytics, Exploratory Analysis, Mechanistic Analysis: Regression. **Statistics:** Understanding attributes and their types, Types of attributes, Discrete and continuous attributes, measuring central tendency: Mean, Mode, Median, measuring dispersion,

Skewness and kurtosis, understanding relationships using covariance and correlation coefficients: Pearson's correlation coefficient, Spearman's rank correlation coefficient, collecting samples, performing parametric tests. **Descriptive Statistics:** Understanding statistics, distribution functions uniform distribution, normal distribution, exponential distribution, binomial distribution. Cumulative distribution function, descriptive statistics.

UNIT–III

NumPy and pandas: Understanding NumPy arrays, NumPy array numerical data types, manipulating array shapes, The stacking of NumPy arrays, Partitioning NumPy arrays, Changing the data type of NumPy arrays, Creating NumPy views and copies, Slicing NumPy arrays, Boolean and fancy indexing, Broadcasting arrays, Creating pandas Data Frames, understanding pandas Series, Reading and querying the Quandl data, describing pandas Data Frames, Grouping and joining pandas Data Frame, working with missing values, creating pivot tables, dealing with dates.

UNIT–IV

Data Cleaning: Exploring data, Filtering data to weed out the noise, Column-wise filtration, row-wise filtration. Handling missing values, dropping missing values, Filling in a missing value, Handling outliers, Feature encoding techniques: one-hot encoding, Label encoding, ordinal encoder. **Correlation:** Introducing correlation, Types of analysis, understanding univariate analysis, understanding bivariate analysis, Understanding multivariate analysis. Discussing multivariate analysis using the Titanic dataset. **Hypothesis Testing and Regression:** Hypothesis testing, Hypothesis testing principle, stats models library, Average reading time, Types of hypothesis testing, T-test.

UNIT–V

Supervised Learning - Regression Analysis, Linear regression, Multiple linear regression, understanding multicollinearity, removing multicollinearity, Dummy variables, developing a linear regression model, Evaluating regression model performance, R-squared, MSE, MAE, RMSE, fitting polynomial regression, Regression models for classification, Logistic regression, Characteristics of the logistic regression model, Types of logistic regression algorithms, Advantages and disadvantages of logistic regression, implementing logistic regression using scikit-learn. **Unsupervised Learning clustering -** PCA and Clustering: Unsupervised learning, reducing the dimensionality of data, PCA, Performing PCA. Finding the number of clusters, the elbow method, the silhouette method, Partitioning data using k-means clustering, Hierarchical clustering, DBSCAN clustering, Spectral Clustering: clustering, Evaluating clustering performance, Internal performance evaluation, The Davies-Bouldin index, The silhouette coefficient, External performance evaluation, The Rand score, The Jaccard score, F-Measure or F1-score, The Fowlkes-Mallows score.

TEXT BOOKS:

1. Chirag Shah, “A Hands-On Introduction to Data Science”, Cambridge University Press, 2020, 978-1108472449.
2. Avinash Navlani, Armando Fandango, and Ivan Idris, “Python Data Analysis: Perform Data Collection, Data Processing, Wrangling, Visualization, and Model Building Using Python”, 3rd ed. Packt Publishing, 2021.

SUGGESTED READING:

1. Suresh Kumar Mukhiya and Usman Ahmed, “Hands-On Exploratory Data Analysis with Python”, Packt Publishing, 2020.

ONLINE RESOURCES:

1. https://nptel.ac.in/courses/106106212
2. https://onlinecourses.nptel.ac.in/noc25_cs101

22CSE19

BIG DATA ANALYTICS
(Professional Elective-I)

Instruction	3 Hours per week
Duration of SEE	3 Hours
SEE	60 Marks
CIE	40 Marks
Credits	3

PREREQUISITE: Proficiency in python programming, Database fundamentals

COURSE OBJECTIVES: This course aims to
1. Introduce the importance of Big Data and the role of the Hadoop framework in analyzing large datasets.
2. Gain practical knowledge in working with Hadoop ecosystem tools such as MapReduce,pig, hive, Spark.
3. Provide hands-on experience in using Spark for real-time stream processing.

COURSE OUTCOMES: After the completion of this course, the student will be able to
1. Define and describe the types, characteristics, and challenges of Big Data and the need for its analysis.
2. Analyze the differences between traditional systems and Hadoop and explain the functions of core components like HDFS and YARN.
3. Develop MapReduce programs and analyze different join strategies and pipeline models for big data processing.
4. Design and execute queries using Hive and Pig to extract, transform, and analyze large datasets.
5. Implement scalable data processing applications using Apache Spark and evaluate its performance over traditional frameworks

CO-PO Articulation Matrix

PO/PSO CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO 1	2	3	1	3	3	3	-	3	3	2	3	2	2	-
CO 2	3	3	3	3	3	2	-	3	3	2	3	2	2	2
CO 3	3	3	3	3	3	2	-	3	3	2	3	2	2	2
CO 4	3	3	3	3	3	2	-	3	3	2	3	1	2	3
CO 5	3	3	3	3	3	3	-	3	3	2	3	2	3	3

1 - Slightly, 2 - Moderately, 3 – Substantially

UNIT-I

Introduction to Big Data: Data and its types: Unstructured, Semi-structured, Structured – Sources of data – Evolution and Definition of Big Data – Characteristics(3Vs/5Vs) and Challenges – Need for Big Data, , Big data integration process, Applications.Overview of Business Intelligence, Data Science, Analytics – Typical analytical architecture – Types of Analytics (Descriptive, Predictive, Prescriptive)

UNIT-II

Hadoop Ecosystem: Requirement of Hadoop Framework - Design principle of Hadoop –Comparison with other system (SQL, RDBMS) - Core ccomponents of Hadoop – Architecture -Hadoop 1 vs Hadoop 2 – HDFS operations, Data ingestion layer, ETL and ELT, Ingestion tools in Hadoop ecosystem, Data ingestion types.

UNIT-III

Big Data Frame Works: Hadoop Framework: Introduction to MapReduce, Processing data with Hadoop using MapReduce, Map Reduce Programming: I/O formats, Map side join-Reduce Side Join,

Secondary Sorting-Pipelining MapReduce jobs. Introduction to YARN, Architecture, Managing Resources and Applications with Hadoop YARN.

UNIT-IV

Hive and Pig: Introduction to Hive: What is Hive, Hive Architecture, Hive data types, Hive file formats, Hive Query Language (HQL), RC File implementation, User Defined Function (UDF).
Introduction to Pig: What is Pig, Anatomy of Pig, Pig on Hadoop, Data types in Pig, Running Pig, Execution Modes of Pig, HDFS Commands, Relational Operators, Eval Function, Complex Data Types, Piggy Bank, User Defined Function, Pig Vs Hive.

UNIT-V

Apache Spark Framework: Introduction to Apache spark-Design principles, Advantages and Disadvantages, Layered architecture, Programming with RDDs: Create RDD- Spark operations - transformations, actions-DAG in Apache spark.
Case Study: Analyze student academic data to find trends, subject difficulty levels, and top performers using big data tools.

TEXT BOOKS:

1. S. Chandramouli, 'Big Data Analytics', University press,2024 edition.
2. Seema Acharya, Subhashini Chellappan, "Big Data and Analytics", Wiley India Pvt. Ltd., 2016.

SUGGESTED READING:

1. Tom White, 'Hadoop: The Definitive Guide', 4th Edition, O'Reilly Media Inc, 2015.
2. Thilinagunarathne, 'Hadoop MapReduce v2 Cookbook', 2nd Edition, Packet Publishing, 2015.
3. Chuck Lam, Mark Davis, Ajit Gaddam, 'Hadoop in Action', Manning Publications, 2016.
4. Alan Gates, 'Programming Pig', O'Reilly Media Inc, 2011.

ONLINE RESOURCES:

1. http://www.planetcassandra.org/what-is-nosql
2. http://www.iitr.ac.in/media/facspace/patelfec/16Bit/index.html
3. https://class.coursera.org/datasci-001/lecture
4. http://bigdatauniversity.com

22MEO01

PRINCIPLES OF DESIGN THINKING
(Open Elective-I)

Instruction	3 L Hours per week
Duration of SEE	3 Hours
SEE	60 Marks
CIE	40 Marks
Credits	3

PREREQUISITE: Nil

COURSE OBJECTIVES: This course aims to
1. Create awareness of design thinking approaches
2. Identify a systematic approach for defining/identifying a problem
3. Create design thinking teams and conduct design thinking sessions collaboratively
4. Apply both critical thinking and design thinking in parallel to solve problems
5. Motivate to apply design thinking concepts to their real life scenarios

COURSE OUTCOMES: After the completion of this course, the student will be able to
1. Understand design thinking and its phases as a tool of innovation
2. Empathize on the needs of the users
3. Define the problems for stimulating ideation
4. Ideate on problems to propose solutions by working as a design thinking team
5. Prototype and test the proposed solutions focusing on local or global societal problems

CO-PO Articulation Matrix

PO/PSO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO 1	1	1	1	1	1	2	2	2	2	2	2	-	-	-
CO 2	1	1	2	1	2	2	2	1	2	2	2	-	-	-
CO 3	1	1	2	2	1	2	2	1	2	2	1	-	-	-
CO 4	2	1	2	2	1	2	2	1	2	2	2	-	-	-
CO 5	2	1	2	2	1	2	2	1	2	2	2	-	-	-

1 - Slightly, 2 - Moderately, 3 – Substantially

UNIT – I

Introduction to Engineering & Thinking: Impact of science/engineering. Thinking and behaviour; Types of thinking – Linear thinking, lateral thinking, design thinking. **Introduction to Design Thinking**: Importance of Design Thinking & Human centric approach – Phases in design thinking process, five-stage model as iterative method, applications of design thinking in various domains.

UNIT – II

Empathize phase: Understanding the unique needs of the user, empathize with the users, steps in empathize phase, developing empathy towards people, assuming a beginner's mind-set (what? why?), steps in immersion activity, body storming.

UNIT – III

Define phase: Define the problem and interpret the result, analysis and synthesis, Personas – Four different perspectives on Personas, steps to creating personas, problem statement, affinity diagrams, empathy mapping.

UNIT – IV

Ideation phase: What is ideation, need, uses, ideation methods; Brainstorming, rules for brainstorming; Mind maps, guidelines to create mind maps; Ideation games; Six Thinking Hats; use of doodling in expressing creative ideas.

UNIT – V

Prototyping phase: Types of prototyping, guidelines for prototyping, storytelling, characteristics of good stories, reaching users through stories, importance of prototyping in design thinking; guidelines to write value proposition. **Testing phase**: Necessity to test, user feedback, conducting a user test, how to test, desirable, feasible and viable solutions, iterate phase.

TEXT BOOKS:

1. T. Brown, "Change by Design: How Design Thinking Creates New Alternatives for Business and Society," 1st ed., HarperCollins, 2009.
2. M. G. Luchs, S. Swan, and A. Griffin, "Design Thinking: New Product Development Essentials from the PDMA," John Wiley & Sons, 2015.
3. P. Soni, "Design Your Thinking: The Mindsets, Toolsets and Skill Sets for Creative Problem-Solving," Penguin Random House India Pvt. Ltd., 2020.

SUGGESTED READING:

1. J. Liedtka, A. King, and K. Bennett, "Solving Problems with Design Thinking: Ten Stories of What Works," Columbia University Press, 2013.
2. B. Ramadurai, "Karmic Design Thinking: A Buddhism-Inspired Method to Help Create Human-Centered Products & Services," 1st ed., 2020.

22ITC11N

COMPUTER NETWORKS LAB

Instruction	3 P Hours per Week
Duration of SEE	3 Hours
SEE	50 Marks
CIE	50 Marks
Credits	1.5

COURSE OBJECTIVES: This course aims to
1. Know about the various network commands.
2. Familiarize with the Configuring Peer to Peer networks and Cisco packet tracer installation.
3. Learn the basic Simulation tools and their installation.
4. Explore the concepts of networks topologies and packet sniffer tool.
5. Acquire knowledge on Socket Programming and SMTP Protocol.

COURSE OUTCOMES: After the completion of this course, the student will be able to
1. Describe the concepts of Networking commands, Peer to Peer Networks.
2. Implement the Cisco Router and the VLAN.
3. Install the various Simulation Tools in Networks.
4. Solve the Network Problems by using Simulators.
5. Implementing Socket Programming and SMTP protocol.

CO-PO Articulation Matrix

PO/PSO	PO	PO	PO	PO	PO	PO	PO	PO	PO	PO	PO	PSO	PSO	PSO
CO	1	2	3	4	5	6	7	8	9	10	11	1	2	3
CO 1	-	2	1	1	-	-	-	-	-	-	-	2	3	3
CO 2	3	2	3	2	1	-	-	-	-	-	-	3	3	2
CO 3	3	1	1	1	1	-	-	-	-	-	-	3	3	3
CO 4	3	2	3	2	1	-	-	-	-	-	-	3	3	2
CO 5	2	2	2	2	1	2	0	2	3	3	3	2	3	3

1 - Slightly, 2 - Moderately, 3 – Substantially

LIST OF EXPERIMENTS:
1. Write and analyze the output of various Network commands such as ping, ipconfig, arp, netstat, tracert, nslookup, hostname, system info etc.,
2. Configure Peer to Peer Network with at least three hosts.
3. Implement Cyclic Redundancy Check method.
4. Configuration of Cisco Router and VLAN.
5. Installation setup of Network simulator software (NS2/NS3/ NetSim /OPNET/ QualNet/ OMNet++ / J-Sim and Cisco Packet Tracer).
6. Simulation of Star topology.
7. Simulation of Stop and Wait Protocol,
8. Simulation of Sliding Window Protocol
9. Simulation of the Routing algorithms (Link State Routing/Distance Vector Routing)
10. Use Wireshark Packet sniffer software and captures TCP, UDP, IP, ARP, ICMP, Telnet, FTP packets
11. Implement Socket Programming.
12. Implement SMTP protocol.

TEXT BOOKS:
1. Andrew S. Tanenbaum, "Computer Networks", Pearson Education,6th Edition,2021.
2. Michael Gregg, "Build Your Own Security Lab", Wiley Publishing, Inc., 2008.
3. Michael E Whitman, Herbert J. Mattord, Andrew Green, "Hands on Information Security lab manual", Cengage Learning, Fourth edition, December 27, 2013.

SUGGESTED READING:

1. James F. Kurose, Keith W. Ross, "Computer Networking – A Top-Down Approach Featuring the Internet", 8th Edition, Pearson Education, 2022.

ONLINE RESOURCES:

1. https://nmap.org
2. https://www.snort.org
3. https://www.wireshark.org
4. NS2 Projects Tutorials | How to install NS2 Software | Network Simulation Tools
5. Network Simulator 2 (NS2) : Steps For Installing NS2 (tutorialsweb.com)
6. The Network Simulator ns-2: Documentation (isi.edu)

22CSC18N

OPERATING SYSTEMS LAB

Instruction	3 P Hours per Week
Duration of SEE	3 Hours
SEE	50 Marks
CIE	50 Marks
Credits	1.5

PREREQUISITE: Operating systems, Programming for problem solving.

COURSE OBJECTIVES: This course aims to
1. Explore Unix/Linux operating system.
2. Analyze various system calls available in Linux/Unix.

COURSE OUTCOMES: After the completion of this course, the student will be able to
1. Understand Linux/Unix environment.
2. Identify and interpret various system programs.
3. Understand and implement shell programming.
4. Simulate memory management, file allocation techniques and process schedules.
5. Analyze process and file management system calls by creating and/or modifying concurrent programs.

CO-PO Articulation Matrix

PO/PSO CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO 1	-	2	-	1	2	1	2	2	2	-	2	2	1	1
CO 2	-	1	1	-	2	2	2	2	2	1	2	2	1	1
CO 3	1	1	1	-	1	2	2	1	2	2	1	1	1	-
CO 4	1	2	2	2	2	1	1	1	1	2	2	1	2	2
CO 5	1	1	-	2	2	1	1	2	2	1	2	1	2	2

1 - Slightly, 2 - Moderately, 3 – Substantially

LIST OF EXPERIMENTS:
1. Demonstration of Linux/Unix file related system calls: mkdir, link, unlink, mount, unmount, users+, chown, chmod, open, close, read, write, lseek, stat, sync.
2. Demonstration of Linux/Unix process related system calls: fork, wait, exec, exit, getpid, getuid, setuid brk, nice, sleep.
3. Shell programming.
4. Implement CPU scheduling algorithms (a) Round Robin (b) SJF (c) FCFS.
5. Implement page replacement algorithms (a) FIFO (b) LRU.
6. Programs to illustrate threads.
7. Demonstration of GNU/Linux IPC mechanisms- Pipes, Semaphores, Shared memory, Message Queues.
8. Implementation of Classical Problems for synchronization (Dining philosopher problem and Producer- Consumer problem).
9. Implementation of Bankers algorithm for Deadlock detection and avoidance.
10. Implementation of Linked, Indexed and Contiguous file allocation methods.

TEXT BOOKS:
1. Galvin, Silberschatz, “Operating System Concepts”, 10th Edition, John Wiley & Sons, 2018.
2. Dhananjay Dhamdhare, “Operating Systems-A Concept Based Approach”, 3rd Edition, McGraw Hill Education, 2017.

SUGGESTED READING:

1. Ekta Walia, “Operating System Concepts”, Khanna Book Publishing, 2020.
2. William Stallings, “Operating Systems Internals and Design Principles”, Pearson Ed., 2012.
3. Charles Crowley, “Operating Systems –A Design Oriented Approach”, McGraw Hill Education, 2017.
4. Andrew S. Tanenbaum, Albert S Woodhull, “Operating systems Design and Implementation”, Pearson Ed., 2009.

ONLINE RESOURCES:

1. https://onlinecourses.nptel.ac.in/noc21_cs88/preview
2. https://onlinecourses.swayam2.ac.in/aic20_sp05/preview

22CSC49

SOFTWARE ENGINEERING LAB

Instruction	2 P Hours per week
Duration of SEE	3 Hours
SEE	50 Marks
CIE	50 Marks
Credits	1

PREREQUISITE: Object Oriented Programming, Software Engineering.

COURSE OBJECTIVES: This course aims to
1. Identify Project Scope, Objectives and infrastructure.
2. Understand Software Engineering methodologies for project development
3. Gain knowledge about Computer Aided Software Engineering (CASE) tools.
4. Use effective communication and technical skills for building quality software.

COURSE OUTCOMES: After the completion of this course, the student will be able to
1. Identify the problem scope and constraints in the problem.
2. Prepare software requirements specifications (SRS) for the system according to standards.
3. Apply the design notations of a structured approach to develop Data Flow Diagrams.
4. Apply/Use the design notations of UML diagrams.
5. Analyze and prepare the documentation for the proposed system.

CO-PO Articulation Matrix

PO/PSO CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO 1	1	1	2	1	-	-	-	2	3	2	-	1	-	3
CO 2	1	1	2	1	-	-	-	2	3	-	-	1	-	3
CO 3	2	3	3	1	-	-	-	2	2	-	-	1	-	3
CO 4	2	1	2	2	-	-	-	2	3	-	-	1	-	3
CO 5	-	-	2	-	-	-	-	2	2	-	-	1	-	3

1 - Slightly, 2 - Moderately, 3 – Substantially

LIST OF EXPERIMENTS:

Select one large Information System/Approach per each team and device the following:
1. Preparation of Software Requirement Specification Document for a given Case Study.
2. Data Flow Diagrams.
3. Use Case Diagrams.
4. Class Diagrams.
5. Sequence Diagrams.
6. Activity Diagrams.
7. State Chart Diagrams.
8. Component Diagrams.
9. Deployment Diagrams.
10. Given a code snippet representing a simple banking system, reverse engineer a class diagram depicting the classes, attributes, methods, and relationships.

TEXT BOOKS:

1. Grady Booch, James Rumbaugh, Ivar Jacobson: "The Unified Modeling Language User Guide", Pearson Education, 2010.
2. Roger S. Pressman, "Software Engineering - A Practitioners Approach", McGraw Hill, 8th Edition, Pearson Education, India, 2014.

ONLINE RESOURCES:

1. https://archive.nptel.ac.in/courses/106/105/106105087/